

⚡ 5-STAGE AUTONOMOUS AGENT WORKFLOW • 500-5,000 ACRES

Autonomous Visual Crop Defense for Commercial Farms.

Sub-2-second computer vision pathology detection, live weather spore modeling, and automated EPA spray work orders with zero human clicks.

[🔗 \[Launch Sentinel Dashboard \]](#)

[\[View Agronomy Specs \]](#)

⏮ STAGE PIPELINE TELEMETRY

✔ Drone Ingestion • ✔ CV Pathology • ✔ Weather Risk • ✔ EPA Formulation • ✔ Tractor Dispatch

READY FOR RUN

FRAME #8109-PIVOT-C • 6SD: 0.12 cm/px

DEFENSE HAZARD LEVEL 4

SECONDARY NECROSIS • 18.5% AREA

⚠ PHYTOPTHORA INFESTANS (LATE BLIGHT) • 96.4%

Lesion Area: 44.2 cm²

SENSOR: SONY ILX-LR1 61MP F/5.6 • ISO 200 • 1/1600s

SPECTRAL INDICES: NDVI 0.41 (STRESSED) • NDRE 0.22

Pathology Verdict

CRITICAL

Detected Pathogen: Late Blight (P. infestans)

Model Confidence: 96.4%

Field Sector: Pivot 03 • Sub-Quadrant B

Infected Perimeter: 4.8 ACRES BUFFERED

Spore Risk Index (DSI): 74/100 (HIGH HUMIDITY)

EPA PRESCRIPTION SPEC

EPA REG. #50534-188

Chlorothalonil 720 SC

Rate: 1.5 pt / acre in 20.0 gal water

🚚 Send Work Order to Tractor #12

ISOBUS TaskController XML generated • 0 clicks required

ANALYSIS THROUGHPUT

1,200 ac/hr

Real-time drone RTK streaming

INPUT REDUCTION

42.8% Average

Surgically targeted nozzle control

PATHOGEN RESOLUTION

0.8 mm/px

Pre-symptomatic spore detection

FLEET COMPATIBILITY

ISOBUS 11783

John Deere, Case IH, Rogator

ENGINEERED FOR HIGH-VALUE PRODUCE & GRAINS

Built to Eliminate Manual Field Scouting.

Sub-Millimeter Computer Vision

Instant leaf pathology segmentation and necrotic lesion calculation. Identifies early-stage Late Blight, Early Blight (Alternaria), and Black Scurf before visual field emergence.

INFERENCE SPEED

< 1.4 SECONDS

Zero-Click Autonomous Workflow

End-to-end background execution from drone tile upload to tractor work-order dispatch. Live API hooks sync weather micro-climates, compute EPA chemical rates, and transmit VRA prescriptions directly to rig terminals.

OPERATOR CLICKS

0

10x Input ROI

Eliminates defensive blanket-spraying, saving \$30,000+ in chemical inputs per season on a standard 2,500-acre rotation. Prevents pathogen-driven field rot and secondary potato storage decay.

AVERAGE NET SAVINGS

\$34,200 / SEASON

Autonomous Telemetry Protocol Standard

ISO 11783-10 COMPLIANT

STAGE PHASE	DATA INPUT SOURCE	ALGORITHMIC ENGINE	OUTPUT ARTIFACT	EXECUTION SLA
01. CAPTURE	Mavic 3M / Wingtra Gen IV Multispectral	Photogrammetry Radiometric Rectification	Calibrated GeoTIFF Orthomosaics	< 3.0 min
02. SEGMENT	High-Res RGB + RedEdge Tile Vectors	Sentinel-YOLO-Agro v5 (Leaf Lesion Focus)	Polygon Centroids & Pathogen ID	< 1.8 sec/tile
03. WEATHER	On-Farm Davis In-Canopy Weather Probe	Wallin Late Blight Disease Severity Index (DSI)	Sporulation Micro-Zone Vectors	Continuous
04. FORMULATE	EPA Registration DB + Farm Inventory ERP	Deterministic Efficacy & Resistance Solver	Batch Tank Mix + Adjuvant Specs	< 400 ms
05. EXECUTE	Precision Ag Telematics Modem	ISOBUS TaskController Auto-Dispatcher	VRA Prescription to Tractor Cab	< 5.0 sec

COMMERCIAL GROWER PLAN

SAAS LICENSE

Commercial Grower Standard

\$499 / month

Billed annually for commercial farms (\$5,988/yr).

- ✔ Up to 2,500 managed acres
- ✔ Unlimited automated CV scans
- ✔ Deterministic EPA tank-mix engine
- ✔ Tractor work-order dispatch
- ✔ Full ISOBUS & Raven Slingshot telematic bridge

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Requires zero hardware installation • Compatible with standard farm drones

Emergency Field Outbreak Support
Active agronomists on duty 24/7 during potato & pulse emergence periods.

1-800-CROPSCAN-0PS

[Inquire Agronomist Support](#)